Explaining Monthly Variation in Auckland Bus Punctuality

A Seasonal and Operational-Event Explanatory Model

Eddy Jiang

Findings and Solutions

1 Industry Problem and Context

Each month Auckland Transport (AT) publishes one figure for bus punctuality. It is the share of operated trips that leave the first stop between one minute early and five minutes late and reach the last stop no more than five minutes late. Over the period analysed in this study, July 2023 to June 2026 (the Observation Period), this Bus Total averaged 89.7%, fell to 84.3% in March 2024, peaked at 94.0% in September 2025, and finished about four pp higher than it started. AT reports it against annual targets of 87%, 88% and 89% but does not say why it moves.

The lack of an explanation matters because the figure carries decisions it was not built for. The metro optimisation manager at AT has described it as more a measure of operator performance, used to calculate the bonuses and abatements operators are paid (RNZ, 2023).

Three dated events fall inside the Observation Period. The driver shortage was declared over on 16 August 2023 (Auckland Transport, 2023a), the Rail Network Rebuild closed the suburban network in eight windows until January 2026 (KiwiRail, 2022; New Zealand Government, 2026), and fare zones were cut from 14 to 9 on 2 February 2025 (Auckland Transport, 2025a). The problem has two further sides.

First, the problem is a measurement problem as much as an explanatory one. The Office of the Auditor-General found that when every stop is checked, only about 70% of bus stop events across 2022/23 and 2023/24 were punctual, and that 10% of trips counted as punctual at both ends were not punctual in between (OAG, 2025). The measure itself changed on 1 July 2023. For the 2,200 route-months in 2023 for which AT published both definitions, the old first-stop measure averaged 96.9% and the new one 87.9%, a gap larger than the whole improvement recorded over the following three years.

Second, it is an attribution problem. One convincing-looking attribution, a 3 pp gain

credited to the fare reform, turns out to be a statistical artefact. Commuters are affected most where route punctuality is lowest, on School services and in South Auckland, operators are paid on the measure, and drivers are exposed because a variable named after their workforce invites blame (New Zealand Council of Trade Unions, 2023). AT has committed to review its metrics (Lowrie, 2025).

AT also has limited room to act. Public transport in Auckland is funded roughly one third each by Auckland Council, NZTA and fares. The share recovered from fares rose from 22% in 2022/23 to 34% in 2024/25 (Auckland Transport, 2025b), under a national policy that expects public transport to deliver value for money and to draw a larger share of its funding from users (Ministry of Transport, 2024). Buses are run by contracted operators whose payments carry punctuality-linked bonuses and abatements (RNZ, 2023), under a framework moving from the Public Transport Operating Model to the Sustainable Public Transport Framework (Ministry of Transport, n.d.). Stop-level records exist inside AT but are not published, so any method that outsiders can reproduce is limited to monthly aggregates.

Demand is still recovering, at 88.8 million boardings in 2024/25 against a 94.7 million target (Auckland Transport, 2025b). The regional plan aims for 150 million trips a year by 2031 (Auckland Council, 2022; Auckland Transport, 2023b), and the City Rail Link (CRL), opened on 13 September 2026, has reshaped the bus network around it (Auckland Transport, 2026a). Growth on this scale depends on a service people trust to run on time, so the punctuality figure will increasingly be used to judge whether the network is ready for it. A figure that cannot explain its own movements is weak evidence for that judgement.

Business informatics treats this as a data-quality problem as well as a modelling one. Data consumers judge quality by fitness for use, including interpretability and consistency, and not by accuracy alone (Wang & Strong, 1996). The AT figure is accurate by its own definition but falls short on fitness for use, because its definition changed mid-series, its route

population is not stated, and it cannot say why it moved.

Research questions and objectives. The study therefore asks one primary research question (RQ). To what extent do multiple operational events (the driver shortage, rail closures and the fare reform), together with seasonal and holiday factors, explain monthly variation in Auckland bus punctuality over the Observation Period? Sub-question 1 asks which individual factors have a statistically significant relationship with punctuality, and sub-question 2 asks whether adding the events explains more than a seasonal model that ignores them. Objective 1 is to identify the significant factors, Objective 2 to quantify each contribution with its uncertainty and cross-check it against the measurement definition, Objective 3 to compare the model with a seasonal baseline, and Objective 4 to turn the results into recommendations for AT.

2 Research Methodology

The study uses a quantitative, quasi-experimental interrupted time series (ITS) design. Dated events enter a multiple linear regression fitted to 36 months of public data, and the result is compared with a seasonal SARIMA baseline (Kontopantelis et al., 2015; Lopez Bernal et al., 2017). A route-level panel separates differences between routes from changes within them, a minimum detectable effect (MDE) and a three-layer test show how large an effect the data could detect, and the two holiday blocks are merged because their effects could not be told apart. The design also covers missing-data handling, industry constraints and an adoption pathway.

2.1 Research Design and Approach

The research adopts a post-positivist paradigm (Creswell & Creswell, 2018). It treats the published punctuality figure as an imperfect but usable proxy for service reliability and seeks explanations that could be shown to be wrong. The approach is deductive (Saunders et al., 2019), since hypotheses about each factor come from the published literature and are tested against official data. The design is quantitative, explanatory and quasi-experimental, with all events estimated together in one regression.

Five constraints shaped this choice. Without a partner organisation the design uses only public workbooks, which also answers OAG Recommendation 5 on reproducibility. Thirty-six months support about eight parameters, so the specification was fixed in advance, since choosing variables after seeing the data would invalidate the standard errors (Lopez Bernal et al., 2017). There was no budget, so all tools and data are free, and the fifteen-week timeline ruled out a data-sharing agreement or participant ethics review. Finally, the ethics

analysis identified two inference risks, blaming a workforce and crediting a policy with an improvement it did not cause, and both became design controls.

2.2 Data Collection Methods

The study is an analysis of secondary data, using routinely published administrative records (Table 2). No sampling was needed because the study uses every published month and every published route, so the data are the full population rather than a sample of it. Three punctuality-and-reliability workbooks and three boardings workbooks supply the variables, and three Bus Performance Reports are used to audit them and to date route changes. An event register records 36 dated events, each with its source link, of which 14 are coded into the model.

These data suit the research question for four reasons. They are the official series AT reports against its targets, they cover the whole window of the three events, they are published by month and route, and they are public, so the analysis can be reproduced. They also have limits. The measure checks only the first and last stops (OAG, 2025; RNZ, 2023), and 36 months give only three seasonal cycles, too few for seasonal patterns to be estimated precisely (Hyndman & Athanasopoulos, 2021). Boardings are published for scheduled routes only (Auckland Transport, 2026c), the definition changed in July 2023 (Auckland Transport, 2025c), the two AT publications disagree for March and April 2024 because one excludes several days (Auckland Transport, 2025c), and no trip-level or trip-weighted figures are published (OAG, 2025).

Table 2

Data sources, coverage and role in the analysis

Source	Content used	Coverage	Role
Monthly Punctuality and Reliability workbooks, FY2024–FY2026 (Auckland Transport, 2026b)	Bus Total row; route rows by service level and area	36 months; 17,704 route-months, 562 routes	Dependent variable; route panel
Weekday Boardings per Service Hour workbooks (Auckland Transport, 2026c)	Route boardings per scheduled weekday service hour	6,635 route-months, 217 routes	Ridership intensity
Bus Performance Reports, 2023 to March 2025 (Auckland Transport, 2025c)	Old and new measures; operators; route-change log	January 2023 to March 2025	Audit; definitional break; route changes
Event register (36 entries), compiled for this study	Dated events, each linked to an official or media source	Late 2022 to April 2026	Event coding (Table 3)

Note. The raw files were kept unchanged throughout the analysis and are identified by their SHA-256 hashes; every transformation is in the accompanying Colab notebook.

The audit covered completeness, duplication and agreement between AT publications. No month is missing, nothing was imputed, and blank route-month cells (months in which a route did not run) were left blank. Route-months shared by two AT publications agree in full for 2023 (1,104 of 1,104) and in 1,826 of 2,183 cases for 2024, with the differences confined to March and April 2024 (means −0.21 and −0.42 pp) and rounding in July. The Bus Performance Report prints the reason above its table, “1 - 3Mar & 14 - 19Apr excluded”, while the monthly workbook has no such exclusion. This study uses the monthly workbook throughout, because AT reports it against its targets and it is the only source covering all 36 months. Removing both disputed months changes no conclusion (S14, Table 7).

Ridership intensity is the unweighted mean, across routes, of weekday boardings per scheduled service hour. It averaged 21.2 and ranged from 16.5 to 26.0. Each event is a 0/1 indicator equal to 1 in the months it was active, dated from official sources, and the fare reform is a permanent step from February 2025 (Table 3). Calendar effects enter as one holiday block covering December, January and the month containing Good Friday. A four-level season was left out of the primary model because six of the nine summer months sit inside the Christmas period (correlation 0.78), and it is tested as a robustness specification instead.

Table 3

Operational event and calendar coding

Variable	Months coded 1	Months	Source
Driver shortage	July–August 2023 (final phase; declared over 16 August 2023)	2	Auckland Transport (2023a)
Rail closure month	Oct 2023; Dec 2023–Jan 2024; Dec 2024–Jan 2025; Apr–Jun 2025; Dec 2025–Jan 2026; Apr 2026	11	KiwiRail (2022); New Zealand Government (2026)
Fare reform (step)	February 2025 onward (zones 14 → 9)	17	Auckland Transport (2025a)
Route/network change	November 2023 (West); July 2024 (North)	2	Bus Performance Reports
Holiday block	Every December and January, and the month containing Good Friday	9	Calendar

Note. Six of the eleven rail closure months fall in December or January (correlation 0.67). A broader rail coding of 20 months is tested as a robustness check.

2.3 Data Analysis Strategy

The analysis has three parts, labelled A to C in Table 4, followed by robustness checks. Part A answers sub-question 1 and Objectives 1 and 2, Part B answers sub-question 2 and Objective 3, and Part C tests whether the Part A mechanism holds within routes and provides the cross-check required by Objective 2 (see *Research questions and objectives*, Section 1).

Table 4

The three analytical parts and where their results are reported

Part	What it does	Data	Answers	Reported in
A	Explanatory regression of events, holidays, ridership and trend, read through SE, CI, VIF, MDE and a joint F-test	Network series, 36 months	Sub-question 1; Objectives 1 and 2	Findings 1 and 2 (Tables 5 and 7)
B	Seasonal SARIMA baseline without events, compared with the regressions on twelve rolling forecasts	Same network series	Sub-question 2; Objective 3	Finding 3 (Table 8)
C	Route panel with route and month fixed effects, rail-closure DiD, and area comparison with the OAG figures	216 scheduled routes, 6,601 route-months; 562 routes for area means	Whether the Part A mechanism holds within routes; Objective 2 cross-check	Findings 1 and 4 (Table 6; Figures 3 and 6)

Part A uses the explanatory regression.

$$Punctuality_t = \beta_0 + \beta_1 Driver_t + \beta_2 Rail_t + \beta_3 Fare_t + \beta_4 Route_t + \beta_5 Holiday_t + \beta_6 Ridership_t + \beta_7 t + \varepsilon_t$$

The Christmas–New Year and Easter blocks were first specified separately. A test of whether their effects are equal cannot reject equality ($F(1, 27) = 0.07$, $p = .798$), so they are carried as one holiday block, and the model exactly as pre-specified is kept as S1. No

event is significant under either version.

The model is estimated by ordinary least squares (OLS) on 36 months, leaving 28 residual degrees of freedom. Each coefficient is read through one chain, from the standard error to the 95% confidence interval, the VIF and the MDE. At 2.903 standard errors, the MDE is the smallest true effect the design had an 80% chance of detecting. A joint F-test asks whether the five event and calendar terms matter together, since overlap prevents them being told apart one by one. Residual autocorrelation is tested with Durbin–Watson, Ljung–Box and Breusch–Godfrey, and Newey–West (HAC) standard errors are reported beside the OLS ones.

In Part B, SARIMA models with a seasonal term were fitted to the same series, and the best converged model was chosen by AIC. Each candidate was refitted at twelve successive months to forecast one month ahead, from July 2025 to June 2026, and RMSE and MAE were compared with a Diebold–Mariano test on the difference. Part C uses the route panel to estimate the ridership relationship within routes, with route and month fixed effects and standard errors clustered by route, and a difference-in-differences for rail closures that compares routes in the four rail-served areas with North and Waiheke routes. Thirteen alternative specifications and four fare effect-shape variants test whether any conclusion depends on a coding choice.

3 Findings and Analysis

This section reports four findings. The first two answer sub-question 1, the third answers sub-question 2, and the fourth checks the results against the measurement definition, before Section 3.5 compares them with the evaluation metrics.

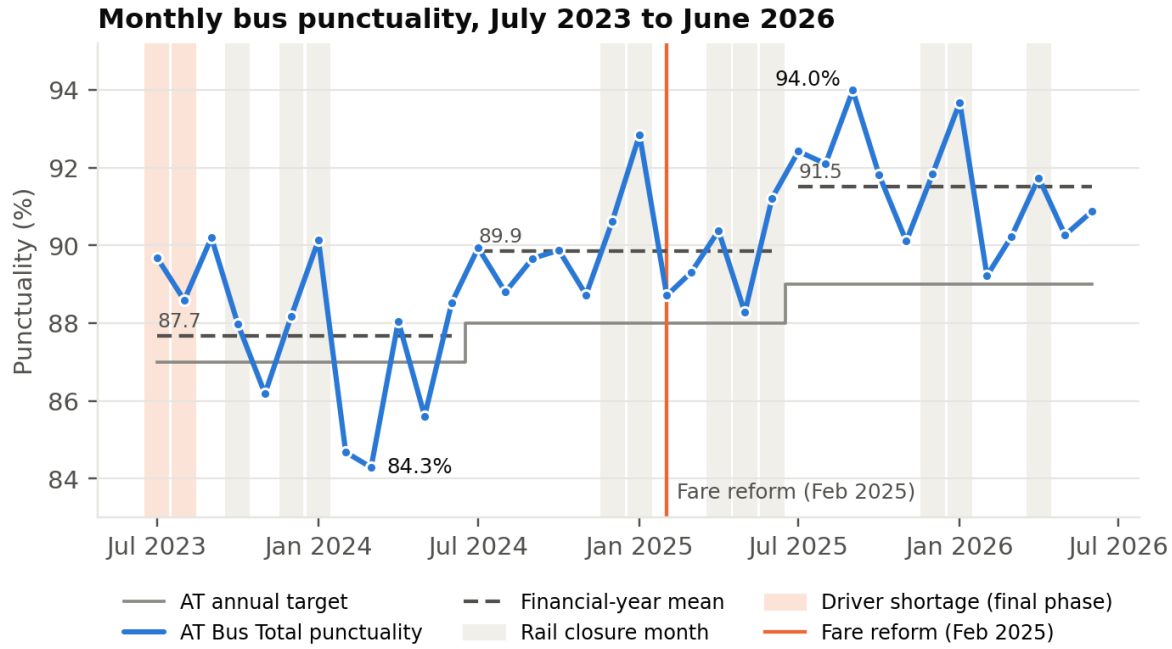
3.1 Finding 1 – Punctuality moves with the calendar and with how busy buses are (Parts A and C)

Mean punctuality rose from 87.7% in FY2024 to 89.9% in FY2025 and 91.5% in

FY2026 (Figure 1). All four months below target occurred in FY2024. An STL decomposition puts the trend at 87.3% at the start and 91.7% at the end, a rise of 4.4 pp, and the calendar pattern spans 5.8 pp. Punctuality is highest in January (+2.5 pp), September (+2.1) and July (+1.8), which contain school holidays, and lowest in February (−2.3), May (−2.2) and March (−2.0), when terms start or run in full (Figure 2).

Figure 1

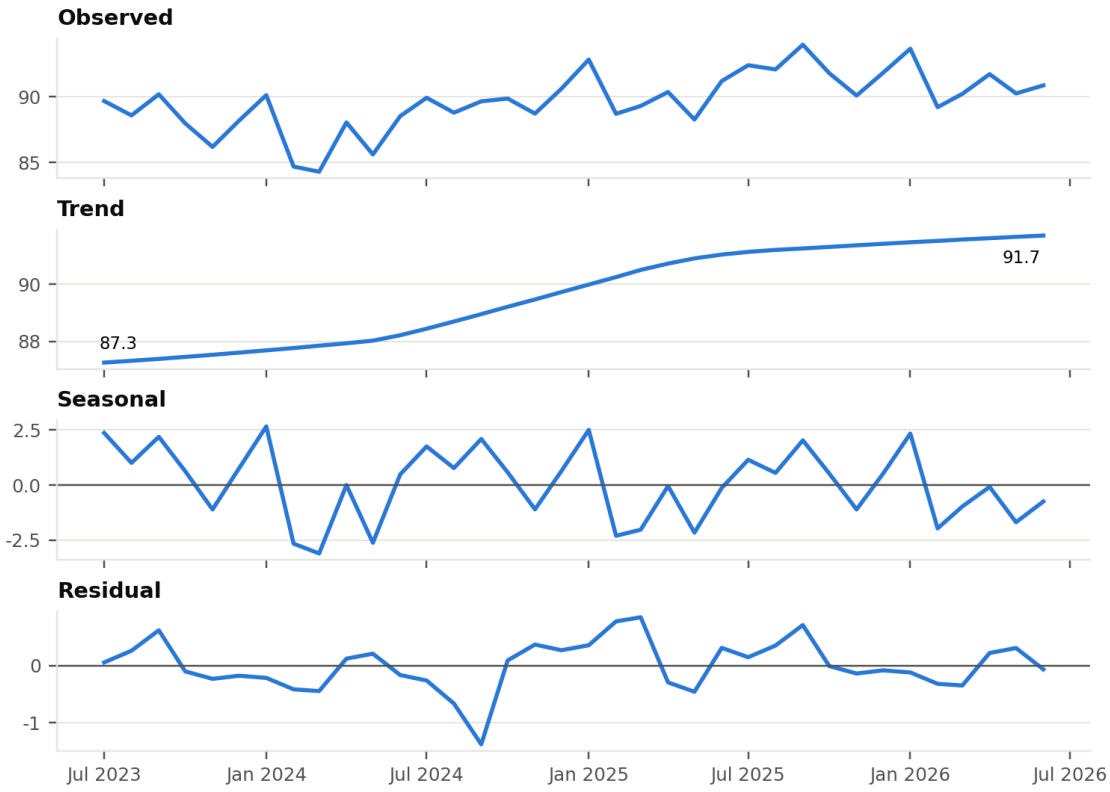
Monthly Bus Total punctuality against the AT annual targets, with event months marked



Note. AT Monthly Punctuality and Reliability workbooks, Bus Total row.

Figure 2

Decomposition of monthly punctuality into trend, calendar pattern and remainder



Note. STL decomposition with a 12-month cycle; components in pp.

The calendar pattern runs mostly through ridership. Calendar month explains 90% of the variation in ridership intensity, and across the twelve calendar months average punctuality and average ridership correlate at −0.86. In the explanatory model each additional boarding per service hour is associated with 0.69 pp lower punctuality in the same month (SE 0.10, $p < .001$, 95% CI −0.88 to −0.49; Table 5). Ridership and the trend alone reach an adjusted R^2 of 0.754, close to the 0.838 obtained with eleven month indicators, and dropping ridership from the full model lowers adjusted R^2 from 0.765 to 0.353. Ridership still matters after calendar month is controlled for (−0.47 pp, $p = .021$), so it is more than a stand-in for the school year. The effect is immediate, as the passenger-activity mechanism predicts (El-Geneidy et al., 2011; Strathman & Hopper, 1993).

Table 5

Explanatory regression: coefficients, uncertainty and detectability

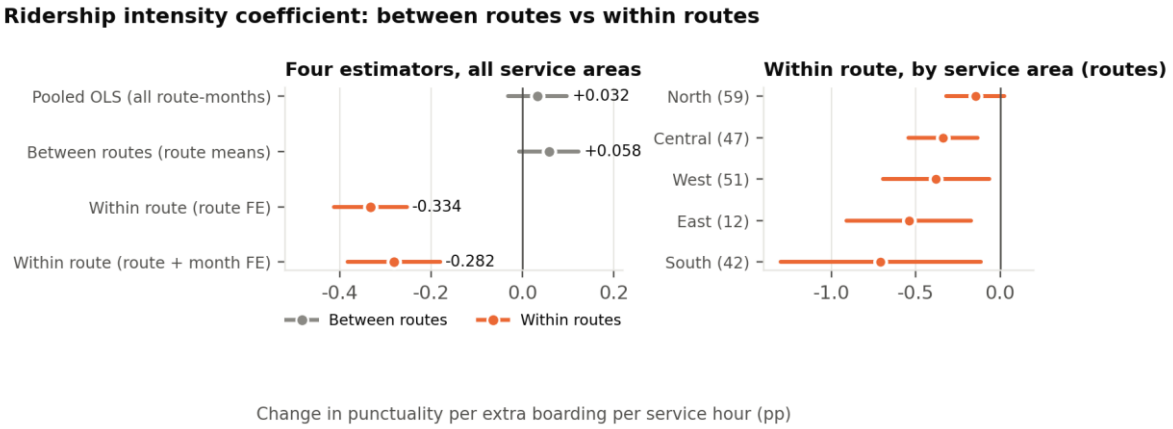
Predictor	Coef. (pp)	SE	p	95% CI	VIF	MDE	HAC p
Driver shortage	1.24	0.93	0.195	[−0.67, 3.15]	1.39	2.71	0.001
Rail closure month	−0.28	0.65	0.669	[−1.61, 1.05]	2.72	1.88	0.604
Fare reform (step)	0.72	0.8	0.377	[−0.92, 2.35]	4.82	2.32	0.242
Route/network change	−0.63	0.85	0.462	[−2.37, 1.10]	1.14	2.46	0.349
Holiday block (merged)	−0.31	0.64	0.637	[−1.63, 1.01]	2.37	1.87	0.476
Ridership intensity	−0.69	0.1	< .001	[−0.88, −0.49]	1.45	0.28	< .001
Linear trend (per month)	0.134	0.04	0.002	[0.05, 0.22]	5.18	0.12	< .001

Note. n = 36, df = 28, adjusted R^2 = 0.765. MDE = 2.903 × SE (80% power, α = .05). HAC p uses Newey–West standard errors with four lags.

The route panel shows why the network figure alone could mislead. Comparing routes with each other, the relationship is slightly positive (+0.03 to +0.06 pp, not significant), plausibly because the busiest routes are Frequent and Rapid services with more bus priority. Within a route the relationship reverses. With route and month fixed effects, a route is 0.28 pp less punctual in months when it carries one more boarding per service hour than its own average (SE 0.052, p < .001, 95% CI −0.38 to −0.18). The effect is negative in all five service areas and largest in the South (−0.71) and East (−0.54; Figure 3).

Figure 3

The same relationship under four estimators, and within routes by service area



Note. Grey bars compare routes with each other; orange bars compare a route with itself. Scheduled routes, 6,601 route-months; 95% intervals clustered by route.

The events do not use this channel. If rail closures pushed train passengers onto buses, closure months would carry more ridership, and controlling for ridership would hide the

closure effect. A three-layer test checked this (Table 6). Closure months carry less ridership, not more, at 19.6 boardings per service hour against 21.9 and −2.47 after adjustment ($p = .044$). Every bias-corrected bootstrap interval for the indirect effect includes zero (Preacher & Hayes, 2008). A route-level difference-in-differences finds no closure effect on ridership in rail-served areas (−0.28, $p = .556$). It does find punctuality 1.14 pp higher there in closure months ($p < .001$), but the gain is concentrated in the three Januaries, at +1.81 pp for holiday closures and +0.34 pp ($p = .419$) for the five closures outside the holidays, which follows summer holiday traffic in the North comparison group rather than the closures.

Table 6

Three-layer test: do events act on punctuality through ridership?

Event	a: event → ridership (p)	Indirect (pp)	Bias-corrected 95% CI	Direct (pp)
Driver shortage	−1.90 (.295)	1.3	[−0.83, 4.49]	1.24
Rail closure month	−2.47 (.044)	1.7	[−0.69, 4.43]	−0.28
Fare reform (step)	+1.43 (.357)	−0.98	[−3.09, 1.29]	0.72
Route/network change	−1.34 (.417)	0.92	[−0.58, 2.14]	−0.63

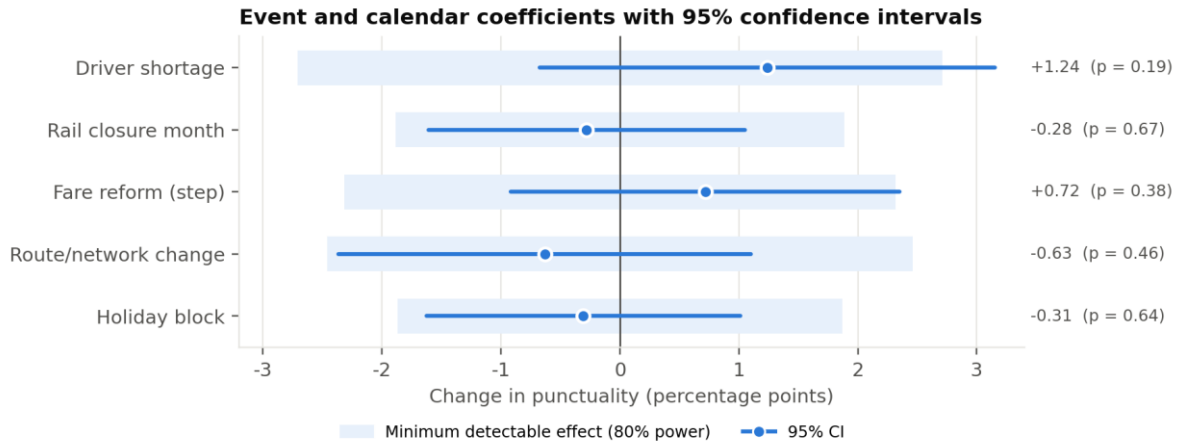
Note. Indirect effect = a × b, with b = −0.69 pp; 3,808 bootstrap resamples of months.

3.2 Finding 2 – No named event is detectable, and the apparent fare effect comes from leaving out the trend (Part A)

None of the five event and calendar terms is significant (Table 5; Figure 4), and together they add almost nothing. The joint test gives $F(5, 28) = 1.31$, $p = .288$, adjusted R^2 rises only from 0.754 to 0.765, and AIC gets worse by 2.4; the model as pre-specified gives $F(6, 27) = 1.07$, $p = .406$. The design could not reliably detect a rail closure effect smaller than 1.9 pp, a fare effect smaller than 2.3 pp or a route-change effect smaller than 2.5 pp. Pooling the three short-lived events into one indicator ($F(2, 28) = 1.59$, $p = .222$) tightens the best MDE to 1.5 pp and still finds nothing (S13, Table 7). The design therefore rules out event effects of the size reported in other transport studies (4–10%, or at least 3.6 pp on a 90% base), but it cannot rule out effects smaller than about 2 pp.

Figure 4

Event and calendar coefficients with 95% confidence intervals and the minimum detectable effect



Note. The shaded band spans ± MDE.

Two coefficients need a closer look, the driver shortage because it rests on only two months, and the fare reform because it looks significant when the trend is left out. The driver shortage covers only two months, so its coefficient (+1.24 pp, HAC p = .001) reflects an estimator failing on two observations and is reported only as context. The trend has a VIF of 5.18, just above the threshold of 5, with the fare step close behind at 4.82, because a step that stays on for 17 months looks almost like a rising line. Removing the driver shortage as well lowers the maximum VIF to 4.46 without making any event significant (S10, Table 7).

The fare reform is the clearest case of a convincing attribution that does not hold. The 17 months after it averaged 90.9% against 88.6% in the 19 months before, a raw gain of 2.4 pp. A model without the trend (S2) credits the reform with +3.05 pp at $p < .001$. With the trend included the coefficient falls to +0.72 pp (p = .377). The trend alone accounts for 74.8% of the variation in the fare step, so without the trend the step absorbs three years of gradual improvement. The improvement after February 2025 is real (step and trend jointly $F(2, 28) = 34.66$, $p < .001$), but these data cannot attribute it to the reform rather than to the improvement already under way.

Four alternative effect shapes were also estimated and all are reported (Lopez Bernal et al., 2017). A one-month lag gives +0.58 pp (p = .471), a two-month lag reverses the sign

(−0.50, p = .539), a three-month pulse gives +1.22 pp (p = .158), and a slope change gives −0.145 pp per month (p = .058). None supports a fare effect. A fare change affects punctuality through boarding (Tirachini, 2013), and this reform changed zones and prices, not how passengers pay, while Table 6 shows it did not change ridership either.

Table 7

Robustness: key coefficients in selected specifications

Specification	df	Adj. R^2	Fare (p)	Rail (p)	Ridership	Max VIF
S9 Holiday blocks merged (primary specification)	28	0.765	+0.72 (.377)	−0.28 (.669)	−0.69	5.18
S1 As pre-specified in the proposal (holiday blocks separate)	27	0.757	+0.67 (.430)	−0.25 (.708)	−0.70	5.38
S2 No trend (artefact demonstration)	29	0.68	+3.05 (< .001)	−0.82 (.272)	−0.71	2.55
S7 Events + trend, no ridership	29	0.353	−0.27 (.840)	+1.42 (.168)	—	5.15
S10 Holiday merged, driver shortage removed	29	0.759	+1.00 (.207)	−0.46 (.484)	−0.71	4.46
S13 Three transient events pooled into one indicator	30	0.755	+0.97 (.226)	—	−0.69	4.77
S14 March and April 2024 removed (disputed between AT publications)	26	0.708	+0.77 (.364)	−0.40 (.593)	−0.66	5.37

Note. Coefficients in pp. Fourteen specifications were estimated; the only one with a significant event term is S2, which omits the trend.

3.3 Finding 3 – Event information adds no forecasting value over a seasonal model (Part B)

The AIC search selected SARIMA$(1,0,0)(1,0,1)_{12}$, which uses only the past of the series. In twelve one-month-ahead forecasts for July 2025 to June 2026 it achieved an RMSE of 0.98 pp and an MAE of 0.81 pp. The full regression achieved 1.37 and 1.16, and the ridership-and-trend regression 1.33 and 1.08 (Table 8; Figure 5). The regressions did worse even though they were given each month's actual ridership. With only twelve forecasts the difference is not statistically significant (Diebold–Mariano t = 1.29, p = .224), so event information did not improve prediction, but SARIMA is not proven better. The baseline is also fragile, with the runner-up only 1.2 AIC points behind.

Table 8

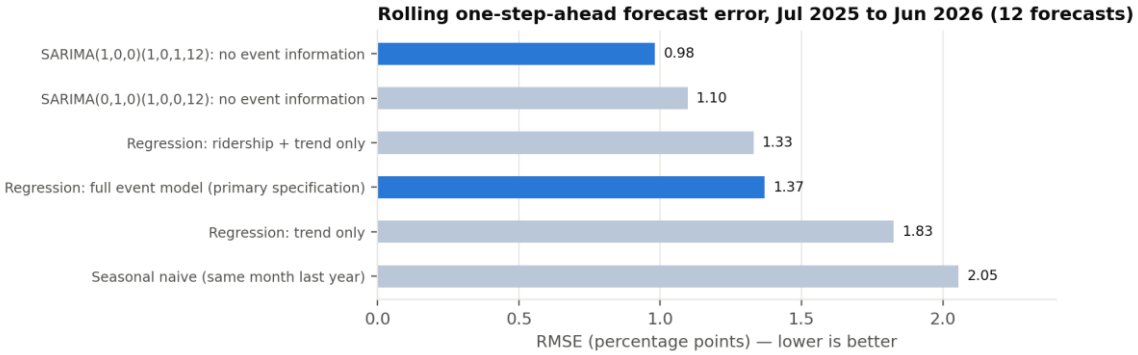
Rolling one-step-ahead forecast accuracy, July 2025 to June 2026 (12 forecasts)

Model	RMSE (pp)	MAE (pp)	Parameters
SARIMA(1,0,0)(1,0,1,12): no event information	0.98	0.81	3
SARIMA(0,1,0)(1,0,0,12): no event information	1.1	0.88	2
Regression: ridership + trend only	1.33	1.08	3
Regression: full event model (primary specification)	1.37	1.16	8
Regression: trend only	1.83	1.56	2
Seasonal naive (same month last year)	2.05	1.72	0

Note. Regression forecasts use actual same-month ridership; SARIMA uses only past punctuality.

Figure 5

Forecast error of the seasonal baseline and the explanatory regressions



Note. Root mean squared error of twelve one-month-ahead forecasts; shorter is better.

3.4 Finding 4 – The network figure hides where and for whom punctuality is poor, and the picture depends on the definition (Part C)

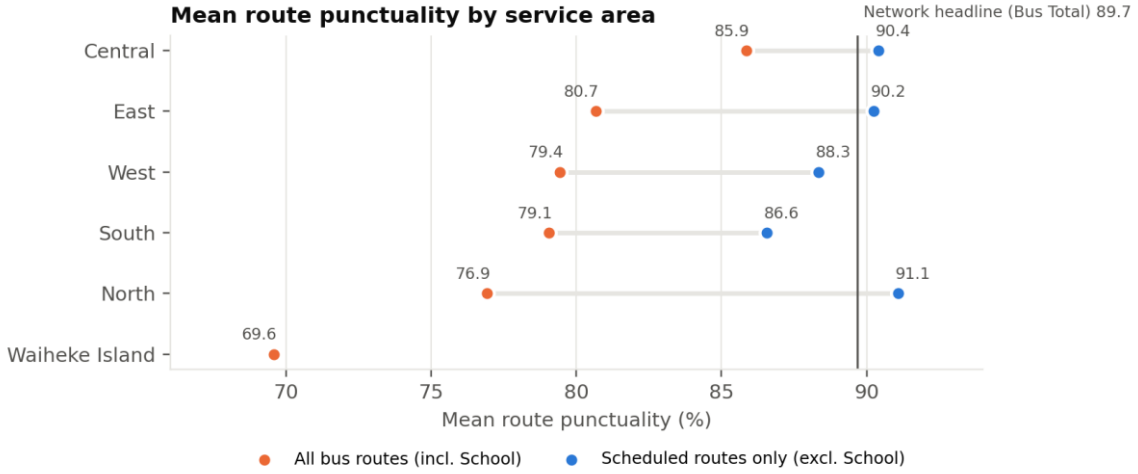
The Bus Total averaged 89.7%, but the mean of published route figures was 79.3%. The gap comes from composition. School services make up 62.7% of published route-months and average 73.2%, the lowest of any service level, but carry few trips, so they barely register in a trip-level headline. Across all routes, area means run from 85.9% in Central Auckland to 69.6% on Waiheke, a spread of 16.3 pp. On scheduled routes alone the spread shrinks to 4.5 pp and the ranking changes (Figure 6). North ranks fifth of six on all routes but first on scheduled routes (91.1%), because 73% of its route-months are School services. South is lowest on scheduled routes (86.6%), so the equity concern for South Auckland survives at a quarter of the headline size.

The OAG all-stop measure gives a third picture (OAG, 2025, Figure 6). Central

Auckland is the least punctual area (62.9%) and Waiheke the most (79.8%), a spread of 16.9 pp, about three times the FY2024 AT scheduled-route spread (5.5 pp). Neither publication states its route population, but the OAG route counts by area match the AT scheduled routes in FY2024. Compared naively with AT all-route means, the OAG figures correlate at −0.76; compared with matched scheduled routes, the correlation is +0.29. Two public and accurate tables cannot be compared without metadata that neither provides, which is the representational failure Wang and Strong (1996) describe.

Figure 6

Mean route punctuality by service area, all routes and scheduled routes



Note. Unweighted means of published route-month figures. Waiheke has fewer than ten scheduled routes, so its scheduled mean is withheld.

3.5 Results against the evaluation metrics

The model meets its explanatory-power and reporting metrics but not the collinearity, residual and alignment ones (Table 9), and each shortfall is traced to a structural feature of the data.

Table 9

Results against the evaluation metrics set at the outset

Criterion (threshold)	Result	Met?
Factor significance and precision (direction, p and 95% CI for every predictor)	Reported for all predictors, with MDE added	Yes
Explanatory power (adjusted $R^2 \geq 0.60$)	Adjusted $R^2 = 0.765$	Yes
Baseline comparison (RMSE and MAE for both models)	Reported; the regression did not beat SARIMA	Reported
Seasonality (STL remainder random)	Ljung–Box $p < .001$	No
Multicollinearity (VIF < 5)	Trend 5.18, fare 4.82; 4.46 in S10	No
Residual autocorrelation (DW 1.5–2.5; Ljung–Box $p > .05$)	DW = 1.69; Ljung–Box $p = .008$; HAC reported	Partly
Robustness (stable without route-change months)	Stable (S8)	Yes
Alignment (≥ 80% of model-flagged months are AT low months)	2 of 3 (67%)	No
Traceability and reproducibility	Every recommendation cites a result; notebook reruns from raw data	Yes

4 Solution Development

The project proposes ABPEM, a reproducible pipeline that turns public AT workbooks into a factor-level explanation of each monthly figure with its uncertainty stated, as well as six recommendations for the AT performance reporting team.

4.1 Proposed Solution / Recommendations

The six recommendations are listed below with the short names used in Table 10.

- **R1 Trend-controlled attribution.** Before attributing any change in punctuality to an intervention, run a trend- and calendar-controlled check and publish the estimate with its confidence interval and MDE, since the 3 pp fare effect without the trend becomes indistinguishable from zero with it.
- **R2 Ridership reporting.** Report ridership intensity beside punctuality by route and area, and use within-route sensitivity to prioritise bus priority and faster boarding such as all-door boarding.
- **R3 Seasonal expectation band.** Publish each month with a seasonal band and describe a month as “changed” only when it falls outside it, because a 5.8 pp seasonal swing makes most month-to-month movement expected.

- **R4 Area and service-level reporting.** Publish punctuality by service area and service level beside the Bus Total, report School services separately, and apply a small-cell rule.
- **R5 Release metadata.** State the definition, route population and any excluded days with every release, which would have made the OAG–AT comparison in Finding 4 possible without reverse-engineering.
- **R6 Separate contractual and customer measures.** Add a customer measure with interim stops at key interchanges, as the auditor suggested (OAG, 2025), which would also address missed connections between late trains and feeder buses (Johnston, 2026).

R1, R3, R4 and R6 follow from the ethics analysis, R2 follows from reporting boardings per service hour, and R5 follows from the data audit. R4 and R5 need little effort, since they re-cut an existing sheet or add a note. R6 is the most consequential and the slowest, because it changes what sets operator payments.

4.2 Implementation Considerations

Table 10

Implementation plan

#	Recommendation	Owner and stakeholders	Resources	Timing	Main obstacle and response
R1	Trend-controlled attribution	AT performance reporting; Council committee	Existing notebook	Next quarterly report	Seen as slowing announcements; apply only to published claims
R2	Ridership reporting	AT reporting; metro optimisation	Existing boardings workbook	Next two releases	Read as blaming passengers; frame as capacity
R3	Seasonal expectation band	AT reporting	Band recomputed annually	Next annual report	Band read as a target; label it an expected range
R4	Area and service-level reporting	AT reporting; local boards; OAG	Existing punctuality sheet	Next monthly release	Exposes weaker areas; publish with caveat
R5	Release metadata	AT data governance	Metadata note per release	Next monthly release	Needs a governance decision; start with a standard template
R6	Separate contractual and customer measures	AT; operators; unions; OAG	Stop-level data AT holds	Next contract round	Changes payments; pilot on interchange routes

Adoption pathway. One analyst in the AT performance reporting team can run the model. The inputs are workbooks AT already publishes each month, the software is open source, and the notebook runs in about a minute, so the recurring cost is roughly a tenth of one role and no new system, licence or data feed is needed. One quarter of parallel running, with the model output beside the current commentary, would show whether the explanations hold before anything is formalised. The suppression rule and the metadata note need a data-governance decision.

ABPEM shows which explanations the data support and how large an effect could have been seen, but it cannot prove that an event had no effect, so it should be presented as a screening check on what may be claimed rather than a causal verdict. Success means an AT analyst can reproduce the coefficients from the raw workbooks within one working day and area figures are published beside the headline.